

Corrections of rainfall data obtained by tipping bucket rain gauge

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ABSTRACT

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Tipping bucket rain gauges are frequently used in local rainfall measurements because of their simplicity and robustness. The accuracy of this type of rain gauge has been tested and a methodology for raw data corrections is presented. In addition to the non-linearity of the sensor, the corrections for “double tipping” and for bucket volume have to be taken into account.

INTRODUCTION

An experimental study has been carried out in order to quantify slight “imperfections” of the tipping bucket sensor. A Rimco R/TBR-8¹ sensor has been tested under both laboratory and field conditions.

It has been realized that this type of sensor has the following three deviations from what is normally expected:

- non-linearity (volume of water disposed by each tip is function of rainfall intensity);
- double tipping after several successive single tips (caused by the difference between the bucket volume and the volume of water supplied to the bucket by the siphon controller, which is also dependent on the rainfall intensity);
- sensitivity of the above two functions on the “adjusted volume” of the bucket.

The results of the laboratory testing are processed and presented in the form suitable for corrections of raw data obtained in field under real storms.

A comprehensive experimental research on tipping bucket performance was carried out by Marsalek (1981) in which various factors influencing the accuracy of the tipping bucket were investigated, such as gauge installation out

¹Manufactured by Analite Pty. Ltd., Melbourne, Australia.

of level, wetting of bucket compartments, as well as the effect of different surface tension of rainwater and tap water used in laboratory. Special attention has been paid to the duration of the bucket movement between two tips as an explanation for underestimated actual intensities by the gauge.

SENSOR FUNCTIONING PRINCIPLES AND CAUSES OF "IMPERFECTIONS"

The principle of the operation of the tipping bucket sensor is shown in Fig. 1. Rainwater collected by the funnel is accumulated in a small reservoir with a siphon type outlet. This device provides intermittent (sequential) dosing of water to the buckets. It is used to reduce the sensitivity of the sensor to the rainfall intensity. The siphon control determines the behaviour of the rain gauge. Volume of water to be disposed by each tip (ΔV) can be adjusted by two screws which are also used for adjustment of equal volumes of left and right bucket compartment. Nominal value of ΔV specified by the manufacturer equals 0.2 mm times the area of the funnel A_f , i.e., it corresponds to the rainfall depth increment of $\Delta h = 0.2$ mm. On the other hand, volume of water disposed to the bucket by the siphon (ΔV_s) depends on the rainfall intensity (i) since some time is required for the collecting reservoir to be emptied by the siphon while the water is pouring continuously into it.

CALIBRATION OF THE RAIN GAUGE

The experiments were carried out in the following steps:

(1) Volumes of left and right bucket compartment are adjusted to be equal using the screws.

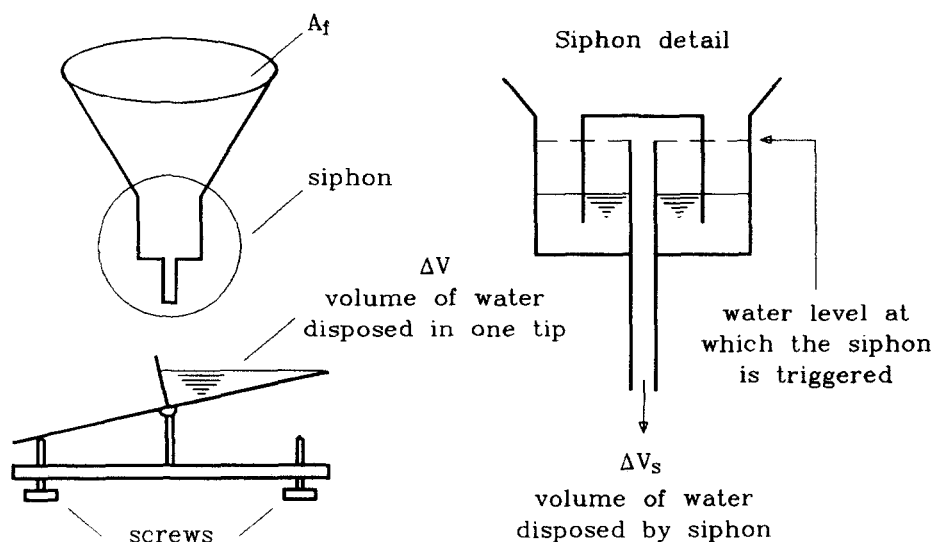


Fig. 1. Principle of tipping bucket operation and explanation of terms.

(2) Static calibration of the bucket volume is performed by careful dropping of water to the compartment by a pipette (ΔV^{stat}).

(3) Start of "rainfall" (constant discharge of water) to allow the funnel to get wet and the bucket to start tipping. This phase should take several bucket tips including at least one double tipping.

(4) Initiation of the stopwatch (coinciding with double tipping).

(5) Counting the number of tips N (30–50). N is divided with the lapsed time T to obtain the number of tips per minute n .

(6) Calculation of ΔV under dynamic conditions (dynamic calibration).

Steps 3 through 6 are repeated several times to obtain the mean value.

The accuracy and performance of the device were tested by varying the following conditions:

- static volume of the compartment (ΔV^{stat} , i.e. the corresponding rainfall depth increment Δh^{stat});
- rainfall intensity;
- siphon present/removed.

RESULTS

Non-linearity of the sensor

The non-linearity of the sensor has been proved earlier (Niemczynowicz, 1986). Assuming linear relationship between rainfall intensity and number of tips per minute, and relying on the nominal value of the water volume necessary for the bucket to tip, leads to erroneous results (about 10% for rainfall intensity of 5 mm/min). Figure 2 presents results of two experiments justifying the non-linear behaviour of the "Rimco" rain gauge. Therefore it is recommended that each tipping-bucket device should undergo the calibration procedure in order to establish non-linear calibration function.

Relationship between rainfall intensity i and number of tips per minute n is obtained by applying the non-linear regression analysis to data obtained by measurements:

$$i = an^b$$

where i is the rainfall intensity, $n = N/T$ is number of tips per minute, T is duration of measurement and a and b are non-linear regression coefficients.

Obtained values for coefficients a and b and the corresponding correlation coefficients are presented in Tables 1 and 2. Graphical presentation of results is given as functions of number of tips per minute (Fig. 3) and volume of one tip (Fig. 4) vs. rainfall intensity for both setups with and without siphon control.

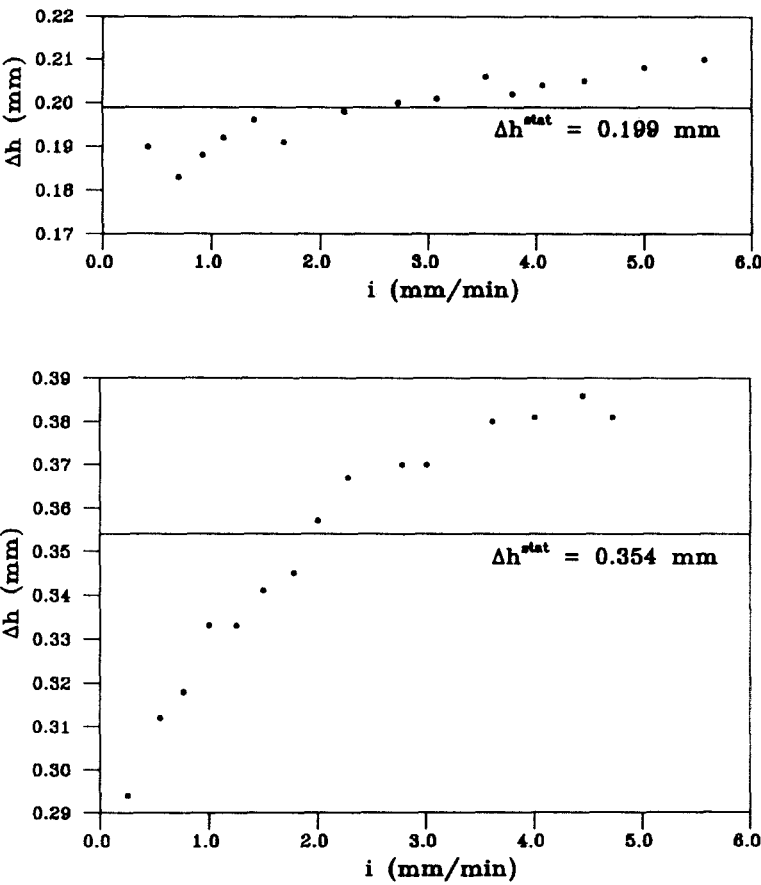


Fig. 2. Non-linear behaviour of the bucket volume for two statically calibrated volumes.

TABLE I

Tipping with siphon control

Δh^{stat} (mm)	a	b	r
0.185	0.144	1.106	0.9996
0.199	0.176	1.050	0.9998
0.252	0.238	1.044	0.9986
0.354	0.294	1.108	0.9999

TABLE 2

Free outflow (siphon removed)

Δh^{stat} (mm)	a	b	r
0.185	0.147	1.092	0.9998
0.200	0.154	1.107	0.9997
0.335	0.261	1.114	0.9993

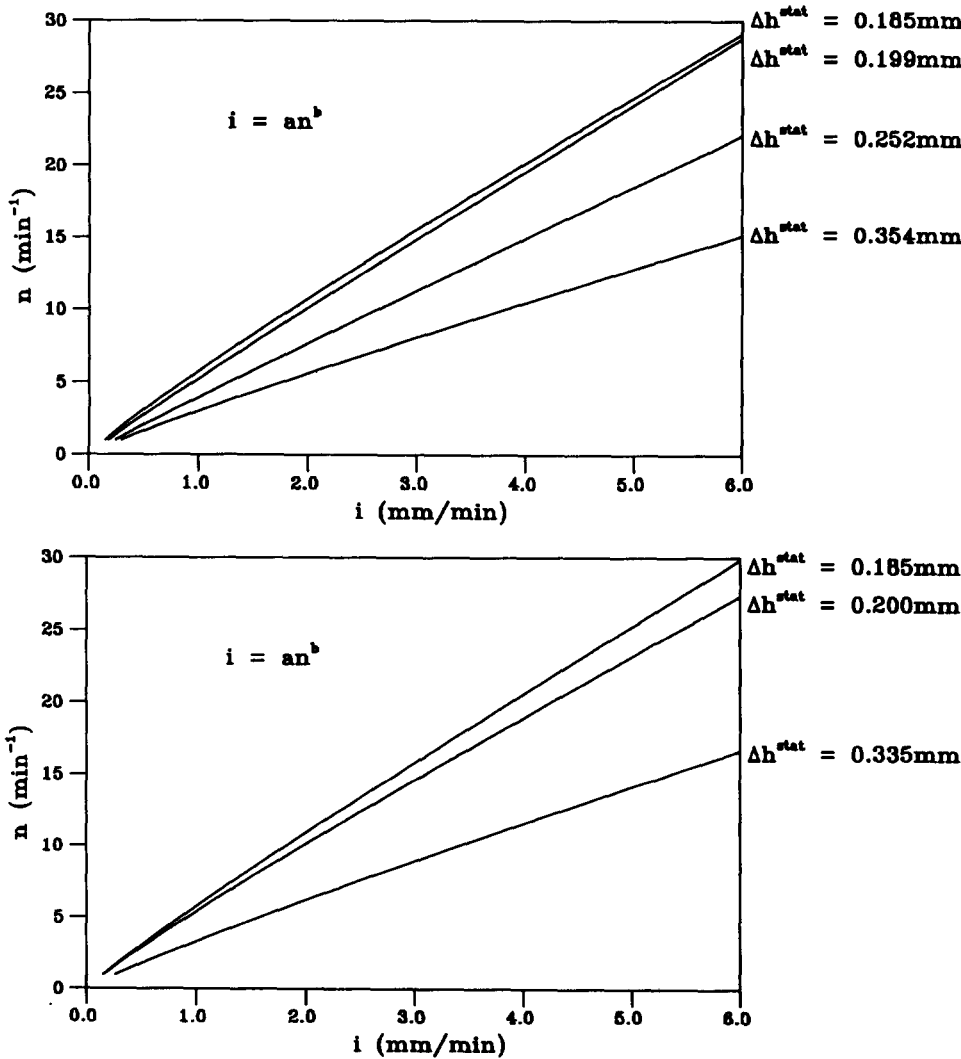


Fig. 3. Number (n) of tips per minute as a function of rainfall intensity and the preset volume of one compartment. On the top: experiments with siphon control. On the bottom: experiments without siphon control.

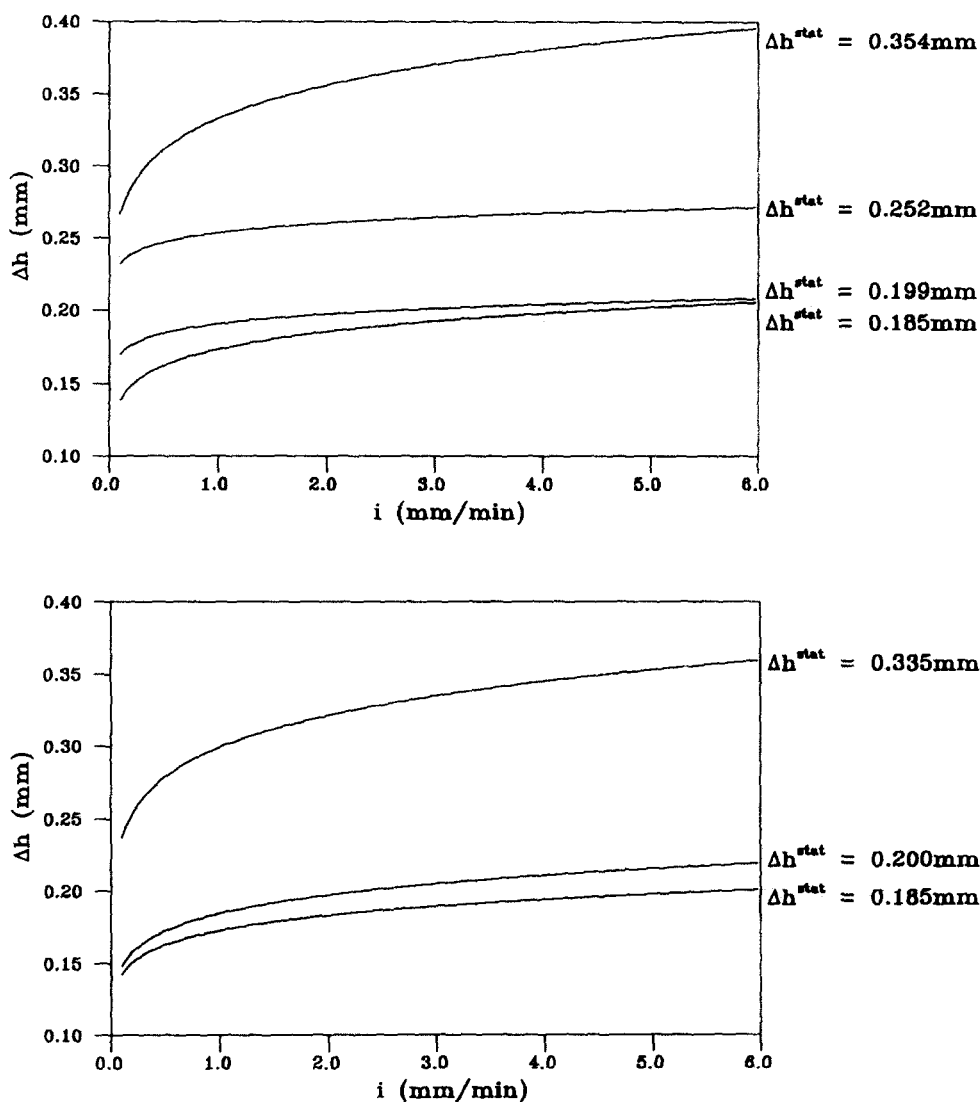


Fig. 4. Rainfall depth increment as a function of rainfall intensity and the preset volume of one compartment. On the top: experiments with siphon control. On the bottom: experiments without siphon control.

Double tipping

The outflow from the siphon (ΔV_s) was tested for a range of intensities and these results are shown in Table 3. Increasing intensity above $i \approx 4$ mm/min causes the continuous outflow from the siphon. It can be seen from Table 3 that the volume of water disposed by the siphon is greater than the nominal

TABLE 3

Outflow from the siphon ΔV_s as a function of rainfall intensity i

i (mm/min)	$\Delta V_s/A_r$ (mm)
0.185	0.280
0.370	0.286
0.555	0.292
0.741	0.304
0.926	0.308
1.000	0.308
1.111	0.309
1.300	0.310
1.481	0.320
1.667	0.335
1.852	0.353
2.000	0.364
2.185	0.378
2.592	0.412
2.778	0.430
3.426	0.490
3.850	0.530

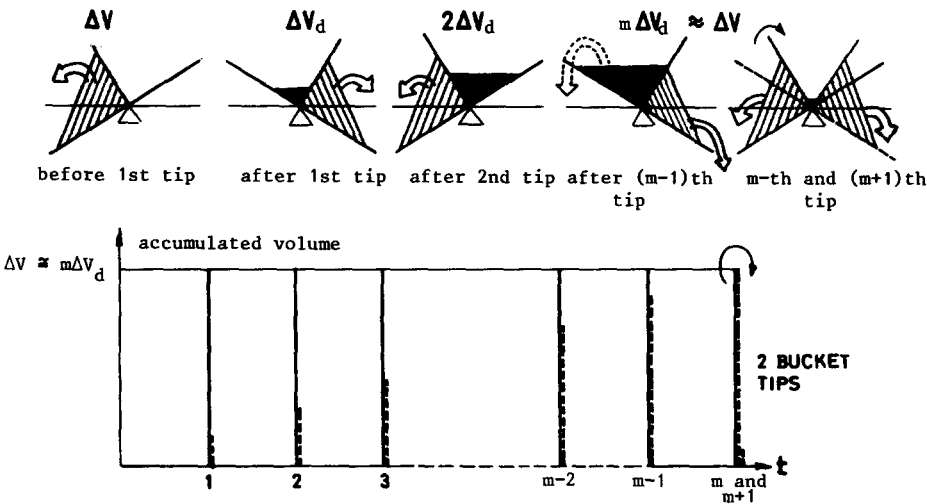


Fig. 5. Schematical presentation of the double tipping effect caused by difference between volume disposed by siphon and volume of one compartment.

bucket volume. After first tip, difference between the two volumes $\Delta V_d = \Delta V_s - \Delta V$ is stored in the opposite compartment and less water is needed for the second tip. After second tip, volume of $2\Delta V_d$ is stored in the opposite

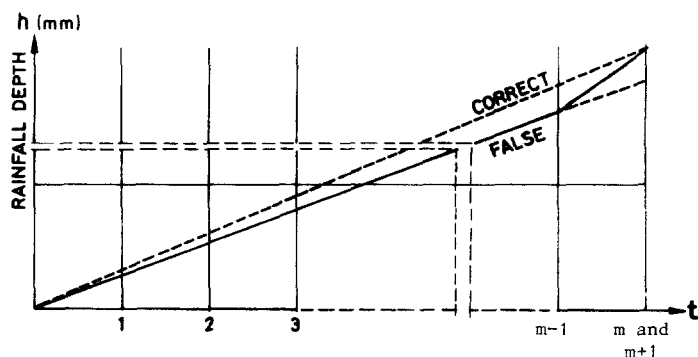


Fig. 6. Effect of double tipping on rainfall pattern (for constant rainfall intensity).

compartment. After several (m) successive tips the accumulated volume reaches the volume sufficient for tipping and the bucket tips twice during that particular emptying of the siphon. This is explained schematically in Fig. 5.

The effect of "double tipping" distorts the actual rainfall pattern in the way shown in Fig. 6. Therefore, if the realistic information on the rainfall pattern is to be obtained, the raw information obtained by recording the time of each tipping (or intervals between two tips) should be corrected. The corrected hyetograph (for constant input intensity) is also shown in Fig. 6.

In the real life cases these corrections have to be made taking into account that the volume of one tipping is dependent on the rainfall intensity. A computer programme has been developed (Bužek, 1986) for the correction of the raw rainfall records obtained by taking into account the "double tipping" and sensor non-linearity effects. The procedure for determining the correction factors has to be repeated regularly during the measurement and sensor's maintenance.

The tipping bucket rain gauge which had been tested in the laboratory was installed on the experimental urban drainage catchment "Miljakovac" in Belgrade and it has been operational for 5 years. Raw data collected on site has been processed and corrected as described earlier. Some general problems of rainfall measurements and data collection at urban drainage experimental catchments are analyzed in the book of Maksimović and Radojković (1986).

CONCLUSIONS

- (1) Tipping bucket is a relatively reliable device for rainfall gauging.
- (2) It is a non-linear device, as the volume (or the corresponding rainfall depth increment) is dependent on the rainfall intensity, and therefore measured data should be corrected.
- (3) If the siphon control is applied, the volume of the emptying of the siphon

is different from the volume of the compartment and it depends on the rainfall intensity. Consequently, measured data should be corrected again.

(4) The above correction procedures should be applied to the raw data by taking into account the real rainfall intensity. The appropriate software has been developed for this purpose.

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